Trackformer 1.1

Causal Western-Pacific Tropical-Cyclone Forecasting

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github.com/yu314-coder/typhoon-predict

Model card and training description

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Abstract

Trackformer 1.1 is a research model for forecasting tropical-cyclone motion, maximum sustained wind, central pressure, radius of maximum wind (RMW), and four-quadrant 34/50/64-kt wind radii over the western Pacific. It combines a causal regional analysis-state route with spatial and temporal structure experts. The model reads observed storm history and analysis information available at the issue time or earlier. It does not consume official agency forecast tracks, positive-lead weather fields, or future observations as inputs. This document describes the released model, its data contract, training procedure, source boundary, and held-out diagnostic comparison with the internal Trackformer 1.2.26 causal baseline.

1 Scope and intended use

The release is an experimental research artifact for reproducible tropical cyclone forecasting and hindcast evaluation. It is not an operational warning system and must not be used for evacuation, aviation, maritime, emergency-management, or other safety-critical decisions. Official meteorological agencies remain the authority for real-world warnings.

2 Forecast contract

The model produces twenty future states at six-hour intervals:

$$\{6, 12, 18, \ldots, 120\} \text{ hours.}$$

Each state contains latitude and longitude, maximum sustained wind, central pressure, RMW, and twelve quadrant radii:

$$[v_{\max}, p_{\mathrm{c}}, r_{\mathrm{mw}}, R34_{NE}, R34_{SE}, R34_{SW}, R34_{NW}, R50_{NE}, \ldots, R64_{NW}].$$

Wind is exposed as a one-minute estimate in knots. Pressure is in hPa. Distances are exposed in kilometres, although the historical training structure labels and the native checkpoint tensors use nautical miles. The public conversion is

$$1 \text{ nautical mile} = 1.852 \text{ km}.$$

The output projection keeps radii non-negative and enforces

$$R34_q \geq R50_q \geq R64_q$$

for every quadrant q.

3 Architecture

3.1 Causal regional route

The route module operates on a western-Pacific analysis domain covering the nearby ocean and the land regions that affect recurvature and decay. It constructs steering views from 850-hPa, 500-hPa, and 200-hPa winds together with 500-hPa height. Inner-flow, deep-layer, ridge, trough, and jet-level views are integrated into a weighted route ensemble. The route sees only analysis states at or before the issue time.

3.2 Intensity and structure experts

The intensity input contract is a nine-step observed storm window with shape 9×54, together with a current four-channel storm-centred analysis patch of shape $4 \times 17 \times 17$. The spatial experts use a transformer encoder for the observed track/environment sequence and a convolutional field encoder for the current patch. Lead queries decode a twenty-step curve for maximum wind and central pressure.

Three structure experts use the same causal history and field representation to estimate RMW and the twelve quadrant radii. The structure decoder is shared across leads but has lead embeddings and a temporal decoder. Expert outputs are averaged and blended using validation-only weights.

3.3 Temporal expert

An additional group of three temporal experts reads the current field and older analysis fields with shape $8 \times 17 \times 17$. It provides a validation-gated residual adjustment to the primary intensity path. The residual branch is disabled when its required causal history is unavailable. It cannot access positive-lead fields.

4 Data and training

4.1 Sources

- **IBTrACS best track:** observed storm position, intensity, RMW, and wind-radius history and future labels for training targets and later verification.
- **Atmospheric analysis/reanalysis:** current, $t - 12$ h, and $t - 24$ h pressure, height, and multi-level wind summaries.
- **SST summaries:** current and past sea-surface-temperature context with availability flags.
- **Static geography:** land/sea state, terrain, land cover, coastline proximity, and geographic location features.

The input policy excludes JMA, JTWC, ECMWF, GFS, GEFS, and other official forecast products. It also excludes positive-lead weather fields and observations that were not available at issue time. Future best-track labels are targets for supervised training or chronological scoring only.

4.2 Chronological storm-disjoint split

The split is determined by each storm's first year:

Split	Storm first year
Training	through 2015
Validation	2016–2019
Test	2020 onward

All normalization statistics, checkpoint selection, and calibration fits are limited to the training and validation periods. The test period is held out until final scoring. Storm identity is kept intact across the split so overlapping windows from one storm cannot cross from training into validation or test.

4.3 Objectives and regularization

The training loss is a masked, lead-weighted Huber/Smooth L1 objective. A temporal curve term penalizes implausible lead-to-lead jumps. Structure channels use explicit availability masks, so an absent historical radius is not interpreted as a zero-radius label. Radius ordering is penalized in the loss and enforced again at the output boundary. Separate kinematic and thermodynamic paths reduce task interference between track and intensity. Three seeds are trained for each expert group and the ensemble/calibration weights are selected on the validation split.

5 Held-out diagnostic comparison

Table 1 reports the final +120 h mean absolute errors. The Trackformer 1.1 values are the released calibrated public path. The Trackformer 1.2.26 values are the causal physics-v4 comparison baseline on 9,994 untouched test windows from storms whose first year is 2020 or later. Lower error is better. The comparison is diagnostic; the 1.2.26 model is not part of this release.

Table 1: Reported held-out +120 h errors.

Metric	Trackformer 1.1	Trackformer 1.2.26	Unit
Maximum wind MAE	20.36	19.88	kt
Central pressure MAE	14.44	13.63	hPa
RMW MAE	25.22	18.53	km
R34 MAE	61.92	46.67	km
R50 MAE	34.24	25.42	km
R64 MAE	20.49	14.77	km

6 Release contents and reproducibility

The GitHub Release archive contains:

- three primary intensity checkpoints;
- three structure checkpoints;
- three temporal checkpoints;

Figure 1: The same comparison as Table 1. The four-quadrant structure values for Trackformer 1.1 were converted from native nautical miles using 1.852 km per nautical mile.

- normalization statistics;
- calibration state and the model manifest.

The main repository contains the public source modules:

- `trackformer_1_1.py` for the public loader;
- `trackformer_1_1_route.py` for the causal route;
- `trackformer_1_1_intensity.py` for intensity and structure;
- `trackformer_1_1_temporal.py` for the temporal expert.

The supported checkpoint format is native PyTorch. GGUF is not an appropriate interchange format for this CNN/Transformer weather model.

7 Limitations

Performance varies with storm structure, basin, analysis source, land interaction, and missing observations. A held-out aggregate metric does not guarantee skill on every individual storm. The release provides no operational warning confidence and must be interpreted with the limitations of the data and causal hindcast design.

License

Code and compact artifacts are released under the repository MIT license. Upstream datasets remain subject to their own terms and attribution requirements.